

What we learned about transient amplification and adaptation (brief report, 2026-09-13)

Summary of the day's three runs of `run_transient_gain` on the paper's network ($n=500$, `celltype_pairs_sfaEI_Sc0p2sig0p1_no` the fast smoke (1 seed, 1-s horizon), the medium run (2 seeds, 2-s horizon, `data/topk_med/transient_gain`, `figs/topk_med`) and a 3-s-horizon run (1 seed, `data/transient_gain_h3`). Full detail and tables: `Transient_gain_frozen_vs_active.md`; the measure and the reasoning behind it: `Non_normal_amplification.md`. Commits 478052a..4061bcf.

The measure

The x-in / x-out transient gain $G(t) = \|P_x \Phi(t) P_x^T\|_2$: a unit perturbation of the dendritic states only, measured on the dendritic states only, with the propagator Φ taken three ways at each sampled state – the dendritic block J_{xx} frozen (the conventional rate-network linearisation), the full Jacobian J frozen (adaptation responds, the state does not move), and active along the trajectory (what a perturbation actually does). The same $n \times n$ block also gives the noise-average gain, the gain along the E/I difference and sum modes and along the leading Lyapunov direction, and the optimal input pattern at the peak.

Six things we learned

1. **The frozen rate-network picture describes a runaway, not a transient.** J_{xx} at a typical state is unstable at about $+6 \text{ s}^{-1}$ in the no- and single-timescale regimes, so $e^{\{J_{xx} t\}}$ grows without bound and its “peak” is the horizon; across states it spans seven decades. The $\omega - \alpha$ margin of the eig-stage figure is the initial slope of that runaway and is $\sim 75 \text{ s}^{-1}$ in every regime, so it cannot summarise adaptation's effect.
2. **The trajectory is more stable than any of its states, in two steps.** Single-timescale regime, medians over 40 states: recurrent block alone $+5.9 \text{ s}^{-1}$, full J with adaptation's feedback $+1.8$, the trajectory itself $+0.48$. Stable regime: $+0.7$, $+0.5$, -0.11 . The first step is adaptation's linear negative feedback at the operating point; the second is motion (the operating point moves away from its own unstable directions), and that second step exists even without adaptation ($+6.0$ vs $+3.35$). This table is the cleanest statement that adaptation belongs in the conceptual model of the recurrent dynamics.
3. **What actually amplifies is 10-20× along a specific pattern, and adaptation changes that only modestly.** Active gain over $e^{\{\lambda_1 t\}}$: $14 / 18 / 6$ for no / single / multiple-timescale adaptation. The decades of difference in the frozen columns are mostly the change in λ_1 ($+4 \rightarrow -0.1$) and in the operating point, not a change in non-normality. Stable regime, the number to quote: a dendritic perturbation is amplified $5.0\times$ [$3.6, 12.4$] at 0.14 s (24 states, 2 seeds), is back to its initial size by 2 s and at half by 3 s, with the isotropic response at 3%. The single-timescale regime has no peak within 3 s: its transient rides on $\lambda_1 > 0$.
4. **The amplifying pattern is not the chaotic direction.** The optimal input is a ~ 70 -neuron E/I-difference pattern with $|\cos| \approx 0.05$ to the leading Lyapunov direction, which lives mostly in the SFA states. Chaos expands along a slow, adaptation-dominated direction; transient amplification is a fast dendritic E/I pattern. The E/I difference mode is the best named direction in every regime (balanced amplification holds), the E/I sum mode is damped below 1 once adapted, and the noise-average gain is ~ 1 : adaptation makes the network selectively sensitive, not globally quiet.
5. **Intermittency is operating-point excursions, not recruited non-normal modes.** States at the onset of a local-Lyapunov-rate excursion carry $\sim 1.5\times$ more worst-case gain than locally contracting states (8.2 vs 5.0 , $p = 0.02$ in the stable regime; 62 vs 42 , $p = 0.09$ single-timescale), but the amplifying direction at an onset is no more aligned with the Lyapunov direction than anywhere else, and the gain along that direction is small (contraction, in the stable regime). The excursion is the state passing through a higher-gain operating point; the divergence happens along a direction a dendritic perturbation barely touches.
6. **For the stimulation idea.** A uniform (E/I sum) push is damped once adaptation is on while the difference pattern amplifies, which supports the prior that the sum mode engages adaptation without a large transient. `find_intermittent_stable` lists the stable-but-intermittent networks to build that analysis on (multiple-timescale regime at $\mu_{IE_relative}$ 2.06 or $\mu_{EE_relative}$ 13.4-15.7; single-timescale at $n = 190$).

What to change in how we talk about it

Keep the $J_{xx} \omega / \alpha$ figure as the conventional view, captioned as the initial slope of the frozen curve. Report the three-propagator curves and the operating-point table (item 2) as the evidence that adaptation is part of the dynamics, not a correction to them. Quote the active peak in the stable regime; in the unstable regimes quote gains at a fixed t together with $e^{\{\lambda_1 t\}}$. Keep the asymptotic rate, the operating-point feedback and the non-normal prefactor apart; the frozen picture conflates them.

Caveats and loose ends

Two seeds (production mode has three); tangent-space linearisation, so every gain is an upper envelope; $\alpha(J)$ via $eigs$ converged on only a third of the stable regime's states (the J -frozen curve's late slope agrees); no 1-s quiet stretch exists in the single-timescale regime, so the onset contrast there is against locally contracting regular states, from a console analysis that the excursion figure does not yet draw. The horizon should be 3 s in the manuscript run.